

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python**-based feature pipelines on **AWS Lambda** and **S3** to support downstream AI model scoring, processing approximately **3,500 events** per second during peak hours.
- Developed and integrated **LLM-powered workflows**, including **RAG pipelines** and agentic patterns, focusing on grounded response generation and state management for enterprise applications.
- Built and deployed **backend services** and **REST APIs** using **FastAPI** to automate AI workflows and facilitate data exchange between AI components and enterprise systems.
- Implemented **AI safety controls** such as input/output validation and guardrails to mitigate prompt injection risks and ensure secure, governed AI system operations.
- Supported **MLOps/LLMOps practices** by integrating **CI/CD pipelines** with **Jenkins** for automated testing, deployment, and monitoring of AI models and applications.
- Instrumented AI systems for quality and latency, collecting user feedback and logging tool-call failures to inform iterative improvements and performance tuning, reducing p95 latency by **180ms**.
- Containerized AI workflow services using **Docker** and deployed them to an **AWS EKS** cluster, configuring basic resource limits and Helm charts for predictable rollouts.

SKILLS

AI/ML Frameworks – LangChain, LlamaIndex, Semantic Kernel, scikit-learn

Programming Languages – Python, Java, TypeScript, SQL

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch, EKS)

Backend Development – Spring Boot, FastAPI, REST APIs, gRPC

AI Concepts – LLM, RAG, Embeddings, Vector Stores, Agentic Workflows, Prompt Management, Hallucination Mitigation, Guardrails

DevOps/MLOps – Docker, Kubernetes, Jenkins, CI/CD, MLOps, LLMOps

PROJECTS

Enterprise AI Workflow Orchestration with LangChain

- Engineered a modular AI workflow system using **LangChain** and **Python** to orchestrate complex tasks involving LLM interactions and external tool usage.
- Implemented a **RAG pipeline** integrating **vector stores** (e.g., FAISS) and hybrid search for retrieving contextually relevant information to ground LLM responses, improving accuracy by **~20%** on a test dataset.
- Developed a **FastAPI** service to expose the AI workflow orchestration capabilities as a reusable API for other enterprise applications.

Secure LLM Integration for Transaction Analysis

- Built a secure API layer using **Java Spring Boot** to interface with an internal LLM service for analyzing financial transaction patterns.
- Integrated AI safety features including **input validation** and **output filtering** to prevent prompt injection and ensure compliance with data privacy regulations, processing approximately **1,500 transactions** daily.
- Deployed the backend service in a **Docker** container on **AWS EC2**, ensuring reliable access for downstream reporting tools.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)